

Uraian deret konvergen dan divergen

No
Date

$$1) \sum_{n=1}^{\infty} \frac{5n}{n^3} = \lim_{n \rightarrow \infty} \frac{5n}{n^3} = \frac{5n+1}{(n+1)^3}$$

$$= \frac{5n}{(n+1)^3} \times \frac{n^3}{n^3} = \frac{5n^3}{n^3 + 3n^2 + 3n + 1}$$

$$= \frac{5 \cdot 1}{1 + 3n^1/n^3 + 3n/n^3 + 1/n^3} = \frac{5}{1} \cdot \frac{1}{\infty} \quad (\text{divergent})$$

$$2) \sum_{n=1}^{\infty} \frac{8^n}{n!} = \lim_{n \rightarrow \infty} \frac{8^n}{n!} = \frac{8^{n+1}}{(n+1)!}$$

$$= \frac{8^{n+1}}{(n+1)!} \times \frac{n!}{8^n} = \frac{8}{n+1} \rightarrow 0 \quad (\text{konvergen})$$

$$3) \sum_{n=1}^{\infty} \frac{n!}{n^{100}} = \lim_{n \rightarrow \infty} \frac{n!}{n^{100}} = \frac{(n+1)!}{(n+1)^{100}}$$

$$= \frac{(n+1)!}{(n+1)^{100}} \times \frac{n^{100}}{n!} = \frac{(n+1)n^{100}}{(n+1)^{100}} = \frac{n^{100}}{(n+1)^{100}} = \frac{n+1}{1} = \infty \quad (\text{tak hingga}) \quad (\text{divergent})$$

$$4.) \sum_{n=1}^{\infty} \frac{s+n}{n!} : \lim_{n \rightarrow \infty} \frac{s+n+1}{(n+1)!} \cdot \frac{n!}{s+n}$$

$$= \frac{s+n+1}{(n+1)!} \cdot \frac{n!}{s+n} = \frac{1}{n+1} \cdot \frac{n}{s+n} = \frac{1/n \cdot 1}{1 + 1/n} = \frac{0}{1} = 0$$

(convergent)